Notes on the non-linear transfer matrix method for optical harmonic generation and mixing in multilayer media

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In these notes, we briefly summarize the nonlinear transfer-matrix method introduced by D. S. Bethune [1] and apply it to a four-layer structure in which the third layer exhibits a second-order nonlinear response.

The linear transfer-matrix method is used to determine the fundamental electric field throughout the multilayer structure. The field in each layer is written as

$$E_j(z,\omega) = E_j^+(\omega)e^{ik_jz} + E_j^-(\omega)e^{-ik_jz}, \tag{1}$$

where E_j^+ and E_j^- are the forward- and backward-propagating amplitudes. The interface and propagation matrices are respectively defined as

$$\mathbf{M}_{ij} = \frac{1}{t_{ij}}\begin{pmatrix} 1 & r_{ij} \\ r_{ij} & 1 \end{pmatrix}, \tag{2}$$

$$\Phi_j = \begin{pmatrix} \phi_j & 0 \\ 0 & \phi_j^{-1} \end{pmatrix}, \qquad \phi_j = e^{ik_jd_j} = e^{in_jk_0d_j}. \tag{3}$$

The amplitudes in any layer are obtained by multiplying the corresponding sequence of interface and propagation matrices, yielding a relation of the form

$$\begin{pmatrix} E_j^+ \\ E_j^- \end{pmatrix} = \mathbf{M}_{j,j-1}\,\mathbf{\Phi}_{j-1}\,\mathbf{M}_{j-1,j-2}\cdots\mathbf{M}_{32}\,\mathbf{\Phi}_2\,\mathbf{M}_{21}\begin{pmatrix} 1 \\ r_{\text{tot}} \end{pmatrix}E_{\text{inc}}. \tag{4}$$

These linear fields constitute the pump fields used in the calculation of the nonlinear source terms.

I. THE NON-LINEAR VERSION

The free-wave fields at frequency $n\omega$ (2ω for SHG anf 3ω for THG) and the source term generated in the nonlinear layer j are related to the fields in the neighboring layers through

$$\mathbf{E}_i = \mathbf{M}_{ij}\mathbf{E}_j + \mathbf{M}_{is}\mathbf{E}_s, \tag{5}$$

$$\mathbf{E}_k = \mathbf{M}_{kj}\mathbf{\Phi}_j\mathbf{E}_j + \mathbf{M}_{ks}\mathbf{\Phi}_s\mathbf{E}_s. \tag{6}$$

From these equations, the field $\mathbf{E}_j$ can be eliminated in order to express the field in any other layer. For simplicity, we consider the neighboring layer k, obtaining

$$\mathbf{E}_k = \mathbf{M}_{kj}\mathbf{\Phi}_j\left(\mathbf{M}_{ji}\mathbf{E}_i + \mathbf{S}_j\right). \tag{7}$$

Here, the source term is defined as

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$$\mathbf{S}_j = \left(\mathbf{\Phi}_j^{-1}\mathbf{M}_{jk}\mathbf{M}_{ks}\mathbf{\Phi}_s - \mathbf{M}_{ji}\mathbf{M}_{is}\right)\mathbf{E}_s, \tag{8}$$

which reduces to Eq. (11b) of Bethune,

$$\mathbf{S}_j = \left(\mathbf{\Phi}_j^{-1}\mathbf{M}_{js}\mathbf{\Phi}_s - \mathbf{M}_{js}\right)\mathbf{E}_s. \tag{9}$$

To derive these expressions, we also used the transfer-matrix property

$$\mathbf{M}_{ij}^{-1} = \mathbf{M}_{ji}. \tag{10}$$

We now relate the fields in the first layer (1) to those in the final layer (f) using Eq. (7),

$$\mathbf{E}_f = \mathbf{M}_{fj}\mathbf{\Phi}_j\left(\mathbf{M}_{j1}\mathbf{E}_1 + \mathbf{S}_j\right). \tag{11}$$

where

$$\mathbf{M}_{fj} = \mathbf{M}_{f(f-1)}\mathbf{\Phi}_{(f-1)}\mathbf{M}_{(f-1)(f-2)}\mathbf{\Phi}_{(f-2)}\cdots\mathbf{M}_{lk}\mathbf{\Phi}_k\mathbf{M}_{kj}, \tag{12}$$

$$\mathbf{M}_{j1} = \mathbf{M}_{ji}\mathbf{\Phi}_i\mathbf{M}_{ih}\mathbf{\Phi}_h\cdots\mathbf{M}_{32}\mathbf{\Phi}_2\mathbf{M}_{21}. \tag{13}$$

Thus, we can write Eq. 12 a of Bethune´s article.

$$\mathbf{R}_{jf}\begin{pmatrix} E_f^+ \\ 0 \end{pmatrix} - \mathbf{L}_{j1}\begin{pmatrix} 0 \\ E_1^- \end{pmatrix} = \begin{pmatrix} S_j^+ \\ S_j^- \end{pmatrix} \tag{14}$$

where $\mathbf{L}_{j1} = \mathbf{M}_{j1}$ is the transfer matrix connecting the first layer to layer j, while $\mathbf{R}_{jf}$ describes the backward propagation from the final layer f to layer j,

$$\begin{aligned}\mathbf{R}_{jf} &= \left(\mathbf{M}_{fj}\mathbf{\Phi}_j\right)^{-1} = \mathbf{\Phi}_j^{-1}\mathbf{M}_{jf} \\ &= \mathbf{\Phi}_j^{-1}\left(\mathbf{M}_{jk}\mathbf{\Phi}_k^{-1}\mathbf{M}_{kl}\mathbf{\Phi}_l^{-1}\cdots\mathbf{M}_{(f-2)(f-1)}\mathbf{\Phi}_{f-1}^{-1}\mathbf{M}_{(f-1)f}\right).\end{aligned} \tag{15}$$

Solving Eq. 14 we get the fields $n\omega$ in the first and final layer:

$$\begin{pmatrix} E_f^+ \\ E_1^- \end{pmatrix} = \frac{1}{R_{11}L_{22} - R_{21}L_{12}}\begin{pmatrix} L_{22} & -L_{12} \\ R_{21} & -R_{11} \end{pmatrix}\begin{pmatrix} S_j^+ \\ S_j^- \end{pmatrix}. \tag{16}$$

II. THE SOURCE FIELD

Now that we have an expression for the fields in the first and final layers, we must determine the source fields $\mathbf{E}_s$. We start from the nonlinear polarization

$$P = \epsilon_0\left(\chi^{(1)}E + \chi^{(2)}E^2 + \chi^{(3)}E^3 + \cdots\right), \tag{17}$$

induced by the fields at frequency ω in layer j. In particular, for second-harmonic generation (SHG), the electric field inside layer j is written as

$$E_j(z,t) = E_j^+ e^{i(k_z z - \omega t)} + E_j^- e^{-i(k_z z + \omega t)}. \tag{18}$$

The corresponding second-order nonlinear polarization is therefore

$$P_2(z,t) = \epsilon_0\chi^{(2)}\left[E_j^+ e^{i(k_z z-\omega t)} + E_j^- e^{-i(k_z z+\omega t)}\right]^2$$
$$= \epsilon_0\chi^{(2)}\left[(E_j^+)^2 e^{i(2k_z z-2\omega t)} + (E_j^-)^2 e^{-i(2k_z z+2\omega t)} + 2E_j^+E_j^- e^{-i2\omega t}\right] \tag{19}$$

This equation can be rewritten by splitting the terms into those that propagate in the positive and negative z-directions, alongside the non-propagating cross-terms:

$$P_2(z,t) = \epsilon_0\chi^{(2)}\left[\begin{pmatrix}(E_j^+)^2 e^{i2n_j(\omega)k_0 z}\\ (E_j^-)^2 e^{-i2n_j(\omega)k_0 z}\end{pmatrix} + \begin{pmatrix}E_j^+E_j^- e^{i2.0.n_j(\omega)k_0 z}\\ E_j^+E_j^- e^{-i2.0.n_j(\omega)k_0 z}\end{pmatrix}\right] e^{-i2\omega t} \tag{20}$$

note that we have replaced the the k_z term by $k_z = n_j(\omega)k_0$ where $n_j(\omega)$ is the refractive index of layer j at the fundamental frequency (ω). The same procedure can be applied to the third non-linear order polarizability, to obtain:

$$P_3(z,t) = \epsilon_0\chi^{(3)}\left[Ae^{i(n_j(\omega)k_0 z-\omega t)} + Be^{-i(n_j(\omega)k_0 z+\omega t)}\right]^3$$
$$= \epsilon_0\chi^{(3)}\left[\begin{pmatrix}A^3 e^{i3n_j(\omega)k_0 z}\\ B^3 e^{-i3n_j(\omega)k_0 z}\end{pmatrix} + \begin{pmatrix}3A^2Be^{in_j(\omega)k_0 z}\\ 3AB^2 e^{-in_j(\omega)k_0 z}\end{pmatrix}\right] e^{-i3\omega t} \tag{21}$$

The nonlinear polarization generates an electric field oscillating at angular frequency 2ω in the case of SHG and at 3ω in the case of THG.

To determine the source electric field, we must solve Eq. (2.1.23) of Boyd's book,

$$\nabla^2\mathbf{E}_n + \frac{\omega_n^2}{c^2}\epsilon^{(1)}(\omega_n)\mathbf{E}_n = -\frac{\omega_n^2}{\epsilon_0 c^2}\mathbf{P}_n(\mathbf{r}), \tag{22}$$

where the nonlinear polarizations for SHG and THG are given in Eqs. (20) and (21), respectively.

As an example, let us consider the first term of Eq. (20). In a one-dimensional geometry, the wave equation reduces to

$$\frac{d^2E_2}{dz^2} + \frac{(2\omega)^2}{c^2}\epsilon_j(2\omega)E_2 = -\frac{(2\omega)^2}{\epsilon_0 c^2}P_2 e^{i2n_j(\omega)k_0 z}. \tag{23}$$

The homogeneous solution corresponds to plane waves of the form

$$E_2^{(h)}(z,t) \propto e^{i[n_j(2\omega)k_0 z-2\omega t]}. \tag{24}$$

To obtain the particular solution, we use the ansatz

$$E_2^{(p)}(z) = Ae^{i2n_j(\omega)k_0 z}. \tag{25}$$

Substituting this expression into Eq. (23), we obtain

$$-4(n_j(\omega))^2k_0^2A + \frac{(2\omega)^2}{c^2}\epsilon_j(2\omega)A = -\frac{(2\omega)^2}{\epsilon_0 c^2}P_2,$$
$$-\frac{(2\omega)^2}{c^2}(n_j(\omega))^2A + \frac{(2\omega)^2}{c^2}\epsilon_j(2\omega)A = -\frac{(2\omega)^2}{\epsilon_0 c^2}P_2. \tag{26}$$

Therefore, the amplitude of the generated source field is

$$A = \frac{P_2}{\epsilon_0\left[(n_j(\omega))^2 - \epsilon_j(2\omega)\right]}. \tag{27}$$

In the derivation above, we used the relation $k_0 = \omega/c$. Furthermore, the dielectric function and refractive index are related through $\epsilon_j(\omega) = [n_j(\omega)]^2$ that can be used to express A either in terms of the dielectric function or the refractive index. Also note that if we use the negative propagation term we will get the same expression for the amplitude of the source field.

We now consider the non-propagating contribution corresponding to the second polarization term in Eq. (20). In this case, the spatial derivative vanishes, leading to

$$\frac{(2\omega)^2}{c^2}\epsilon_j(2\omega)A = -\frac{(2\omega)^2}{\epsilon_0 c^2}P_2. \tag{28}$$

Therefore, the amplitude of the second source field is given by

$$A = -\frac{P_2}{\epsilon_0\epsilon_j(2\omega)}. \tag{29}$$

III. REFLECTION AND TRANSMISSION OF THE SOURCE TERMS

Note that, similarly to the linear case, the nonlinear fields (SHG and THG) must also satisfy the continuity conditions at the interfaces. Therefore, the usual Fresnel reflection and transmission coefficients can still be employed, provided that the appropriate source refractive index is used, namely $n_s = n_j(\omega)$ or $n_s = 0$ in the case of SHG.

Although these coefficients are formally defined for interfaces between different media, Eq. 9 contains the matrix M_{js}, which describes the coupling between the nonlinear source field and the propagating harmonic fields. To evaluate this matrix, we start from the definition

$$\begin{aligned} M_{js} &= M_{jk}M_{ks} && (30)\\ &= \frac{1}{t_{jk}}\begin{pmatrix} 1 & r_{jk} \\ r_{jk} & 1 \end{pmatrix}\frac{1}{t_{ks}}\begin{pmatrix} 1 & r_{ks} \\ r_{ks} & 1 \end{pmatrix} && (31)\\ &= \frac{1}{t_{jk}t_{ks}}\begin{pmatrix} 1 + r_{jk}r_{ks} & r_{jk} + r_{ks} \\ r_{jk} + r_{ks} & 1 + r_{jk}r_{ks} \end{pmatrix}. && (32) \end{aligned}$$

Using

$$r_{ij} = \frac{n_i - n_j}{n_i + n_j}, \tag{33}$$

$$t_{ij} = \frac{2n_i}{n_i + n_j}, \tag{34}$$

we obtain

$$\begin{aligned} 1 + r_{jk}r_{ks} &= 1 + \frac{(n_j - n_k)(n_k - n_s)}{(n_j + n_k)(n_k + n_s)} && (35)\\ &= \frac{(n_j + n_k)(n_k + n_s) + (n_j - n_k)(n_k - n_s)}{(n_j + n_k)(n_k + n_s)} && (36)\\ &= \frac{n_j n_k + n_j n_s + n_k^2 + n_k n_s + n_j n_k - n_j n_s - n_k^2 + n_k n_s}{(n_j + n_k)(n_k + n_s)} && (37)\\ &= \frac{2n_j n_k + 2n_k n_s}{(n_j + n_k)(n_k + n_s)} && (38)\\ &= \frac{2n_k(n_j + n_s)}{(n_j + n_k)(n_k + n_s)}. && (39) \end{aligned}$$

Similarly,

$$r_{jk} + r_{ks} = \frac{n_j - n_k}{n_j + n_k} + \frac{n_k - n_s}{n_k + n_s} \tag{40}$$

$$= \frac{(n_j - n_k)(n_k + n_s) + (n_k - n_s)(n_j + n_k)}{(n_j + n_k)(n_k + n_s)} \tag{41}$$

$$= \frac{n_j n_k + n_j n_s - n_k^2 - n_k n_s + n_j n_k + n_k^2 - n_j n_s - n_k n_s}{(n_j + n_k)(n_k + n_s)} \tag{42}$$

$$= \frac{2n_j n_k - 2n_k n_s}{(n_j + n_k)(n_k + n_s)} \tag{43}$$

$$= \frac{2n_k(n_j - n_s)}{(n_j + n_k)(n_k + n_s)}. \tag{44}$$

Also,

$$t_{jk} t_{ks} = \frac{2n_j}{n_j + n_k} \frac{2n_k}{n_k + n_s} \tag{45}$$

$$= \frac{4n_j n_k}{(n_j + n_k)(n_k + n_s)}. \tag{46}$$

Therefore,

$$M_{js} = \frac{(n_j + n_k)(n_k + n_s)}{4n_j n_k} \begin{pmatrix} \frac{2n_k(n_j + n_s)}{(n_j + n_k)(n_k + n_s)} & \frac{2n_k(n_j - n_s)}{(n_j + n_k)(n_k + n_s)} \\ \frac{2n_k(n_j - n_s)}{(n_j + n_k)(n_k + n_s)} & \frac{2n_k(n_j + n_s)}{(n_j + n_k)(n_k + n_s)} \end{pmatrix} \tag{47}$$

$$= \frac{1}{2n_j} \begin{pmatrix} n_j + n_s & n_j - n_s \\ n_j - n_s & n_j + n_s \end{pmatrix}. \tag{48}$$

The above result essentially shows that the source coupling can be written in terms of a standard transmission matrix with coefficients given by

$$r_{js} = \frac{n_j - n_s}{n_j + n_s}, \qquad t_{js} = \frac{2n_j}{n_j + n_s}, \tag{49}$$

It is important to emphasize that the refractive index n_j is evaluated at the nonlinear frequency, namely 2ω for SHG or 3ω for THG.

IV. SHG IN THE FOUR-LAYER STRUCTURE

We are now in a position to analyze the four-layer structure. The first layer is air, the second is a thin oxide layer with unknown thickness and refractive index, the third is the nonlinear layer, and the fourth is the substrate, which is assumed to be glass. For the nonlinear layer, we assume that the refractive indices at both ω and 2ω are known.

A. Overall Transfer Matrix

To determine the pump field inside the nonlinear layer, we first calculate the overall transfer matrix of the structure at the fundamental frequency,

$$\mathbf{T}(\omega) = \mathbf{M}_{43}\mathbf{\Phi}_3\mathbf{M}_{32}\mathbf{\Phi}_2\mathbf{M}_{21}. \tag{50}$$

Once the overall transfer matrix has been obtained, the corresponding reflection coefficient can be calculated as

$$r = -\frac{T_{21}}{T_{22}}, \tag{51}$$

where T_{21} and T_{22} are the elements of the overall transfer matrix defined in Eq. (50).

B. Field Inside the Nonlinear Layer

Once the reflection coefficient has been determined, the forward- and backward-propagating pump fields inside the nonlinear layer (layer 3) can be obtained by propagating the incident field through the first two interfaces and the oxide layer. The corresponding field amplitudes are given by

$$\begin{pmatrix} E_3^+(\omega) \\ E_3^-(\omega) \end{pmatrix} = \mathbf{M}_{32}\mathbf{\Phi}_2\mathbf{M}_{21} \begin{pmatrix} 1 \\ r \end{pmatrix} E_0, \tag{52}$$

where (E_0) is the amplitude of the incident pump field and (r) is the overall reflection coefficient of the multilayer structure of eq. 51.

C. The R and L matrices

To determine the SHG fields in the first and last layers, we make use of the **R** and **L** matrices introduced in Eqs. (14). In particular, for the interface between layers 3 and 4, the matrix $\mathbf{R}_{34}$ is defined as

$$\mathbf{R}_{34} = \mathbf{\Phi}_3^{-1}\mathbf{M}_{34}, \tag{53}$$

where

$$\mathbf{\Phi}_3^{-1} = \begin{pmatrix} \phi_3^{-1} & 0 \\ 0 & \phi_3 \end{pmatrix}, \qquad \mathbf{M}_{34} = \frac{1}{t_{34}} \begin{pmatrix} 1 & r_{34} \\ r_{34} & 1 \end{pmatrix}. \tag{54}$$

Performing the matrix multiplication explicitly, we obtain

$$\begin{aligned} \mathbf{R}_{34} &= \begin{pmatrix} \phi_3^{-1} & 0 \\ 0 & \phi_3 \end{pmatrix} \left[\frac{1}{t_{34}} \begin{pmatrix} 1 & r_{34} \\ r_{34} & 1 \end{pmatrix} \right] \\ &= \frac{1}{t_{34}} \begin{pmatrix} \phi_3^{-1} & \phi_3^{-1} r_{34} \\ \phi_3 r_{34} & \phi_3 \end{pmatrix}. \end{aligned} \tag{55}$$

where

$$\phi_3 = \exp[i\, n_3(2\omega)\, k_0'\, d_3]\,, \tag{56}$$

$$\phi_3^{-1} = \exp[-i\, n_3(2\omega)\, k_0'\, d_3]\,, \tag{57}$$

with $k_0' = 2\omega/c$. It should be emphasized that these phase factors describe the propagation of free waves at the SHG frequency 2ω.

Using the fundamental frequency ω as a reference, the corresponding phase factors become

$$\phi_3 = \exp[i\, n_3(\omega)\, 2k_0\, d_3]\,, \tag{58}$$

$$\phi_3^{-1} = \exp[-i\, n_3(\omega)\, 2k_0\, d_3]\,, \tag{59}$$

where $k_0 = \omega/c$. Equivalently, one may write the phase accumulation as $\exp[\pm i\, n_3(\omega)(2\omega/c)d_3]$.

We now evaluate the matrix $\mathbf{L}_{31}$, which relates the field amplitudes in layer 3 to those in layer 1. From Eq. (61), this matrix is given by

$$\mathbf{L}_{31} = \mathbf{M}_{32}\mathbf{\Phi}_2\mathbf{M}_{21}, \tag{60}$$

or explicitly,

$$\mathbf{L}_{31} = \left[\frac{1}{t_{32}}\begin{pmatrix} 1 & r_{32} \\ r_{32} & 1 \end{pmatrix}\right]\begin{pmatrix} \phi_2 & 0 \\ 0 & \phi_2^{-1} \end{pmatrix}\left[\frac{1}{t_{21}}\begin{pmatrix} 1 & r_{21} \\ r_{21} & 1 \end{pmatrix}\right]. \tag{61}$$

We first multiply the propagation matrix by the interface matrix at the 2–1 boundary,

$$\begin{aligned} \mathbf{\Phi}_2\mathbf{M}_{21} &= \frac{1}{t_{21}}\begin{pmatrix} \phi_2 & 0 \\ 0 & \phi_2^{-1} \end{pmatrix}\begin{pmatrix} 1 & r_{21} \\ r_{21} & 1 \end{pmatrix} \\ &= \frac{1}{t_{21}}\begin{pmatrix} \phi_2 & \phi_2 r_{21} \\ \phi_2^{-1} r_{21} & \phi_2^{-1} \end{pmatrix}. \end{aligned} \tag{62}$$

Substituting this result into Eq. (61), we obtain

$$\begin{aligned} \mathbf{L}_{31} &= \frac{1}{t_{32}t_{21}}\begin{pmatrix} 1 & r_{32} \\ r_{32} & 1 \end{pmatrix}\begin{pmatrix} \phi_2 & \phi_2 r_{21} \\ \phi_2^{-1} r_{21} & \phi_2^{-1} \end{pmatrix} \\ &= \frac{1}{t_{32}t_{21}}\begin{pmatrix} \phi_2 + r_{32}r_{21}\phi_2^{-1} & r_{21}\phi_2 + r_{32}\phi_2^{-1} \\ r_{32}\phi_2 + r_{21}\phi_2^{-1} & r_{32}r_{21}\phi_2 + \phi_2^{-1} \end{pmatrix}. \end{aligned} \tag{63}$$

Therefore, the transfer matrix connecting layers 1 and 3 can be written as

$$\mathbf{L}_{31} = \frac{1}{t_{32}t_{21}}\begin{pmatrix} \phi_2 + r_{32}r_{21}\phi_2^{-1} & r_{21}\phi_2 + r_{32}\phi_2^{-1} \\ r_{32}\phi_2 + r_{21}\phi_2^{-1} & r_{32}r_{21}\phi_2 + \phi_2^{-1} \end{pmatrix}. \tag{64}$$

where $\phi_2 = \exp[i\, n_2(2\omega)\, 2k_0\, d_2]$ and $\phi_2^{-1} = \exp[-i\, n_2(2\omega)\, 2k_0\, d_2]$.

D. The SHG field in the layer 1 and 4

Using the expressions for the elements of the matrices R and L, Eq. (16) can be written explicitly by substituting $R_{11} = \phi_3^{-1}/t_{34}$ and $R_{21} = \phi_3 r_{34}/t_{34}$ from Eq. (55), together with $L_{22} = (r_{32}r_{21}\phi_2 + \phi_2^{-1})/(t_{32}t_{21})$ and $L_{12} = (r_{21}\phi_2 + r_{32}\phi_2^{-1})/(t_{32}t_{21})$ from Eq. (64). This yields

$$\begin{pmatrix} E_4^+ \\ E_1^- \end{pmatrix} = \frac{1}{D}\begin{pmatrix} \dfrac{r_{32}r_{21}\phi_2 + \phi_2^{-1}}{t_{32}t_{21}} & -\dfrac{r_{21}\phi_2 + r_{32}\phi_2^{-1}}{t_{32}t_{21}} \\ \dfrac{\phi_3 r_{34}}{t_{34}} & -\dfrac{\phi_3^{-1}}{t_{34}} \end{pmatrix}\begin{pmatrix} S_j^+ \\ S_j^- \end{pmatrix}, \tag{65}$$

where

$$\begin{aligned} D &= \frac{\phi_3^{-1}}{t_{34}}\frac{r_{32}r_{21}\phi_2 + \phi_2^{-1}}{t_{32}t_{21}} - \frac{\phi_3 r_{34}}{t_{34}}\frac{r_{21}\phi_2 + r_{32}\phi_2^{-1}}{t_{32}t_{21}} \\ &= \frac{\phi_3^{-1}\left(r_{32}r_{21}\phi_2 + \phi_2^{-1}\right) - \phi_3 r_{34}\left(r_{21}\phi_2 + r_{32}\phi_2^{-1}\right)}{t_{34}t_{32}t_{21}}. \end{aligned} \tag{66}$$

Therefore,

$$\begin{pmatrix} E_4^+ \\ E_1^- \end{pmatrix} = \frac{1}{\phi_3^{-1}\big(r_{32}r_{21}\phi_2 + \phi_2^{-1}\big) - \phi_3 r_{34}\big(r_{21}\phi_2 + r_{32}\phi_2^{-1}\big)} \begin{pmatrix} t_{34}\big(r_{32}r_{21}\phi_2 + \phi_2^{-1}\big) & -t_{34}\big(r_{21}\phi_2 + r_{32}\phi_2^{-1}\big) \\ \phi_3 r_{34}\, t_{32}t_{21} & -\phi_3^{-1} t_{32}t_{21} \end{pmatrix} \begin{pmatrix} S_3^+ \\ S_3^- \end{pmatrix}. \tag{67}$$

Carrying out the matrix–vector multiplication, we obtain

$$E_4^+ = \frac{t_{34}\big(r_{32}r_{21}\phi_2 + \phi_2^{-1}\big)\, S_3^+ - t_{34}\big(r_{21}\phi_2 + r_{32}\phi_2^{-1}\big)\, S_3^-}{\phi_3^{-1}\big(r_{32}r_{21}\phi_2 + \phi_2^{-1}\big) - \phi_3 r_{34}\big(r_{21}\phi_2 + r_{32}\phi_2^{-1}\big)}, \tag{68}$$

$$E_1^- = \frac{\phi_3 r_{34}\, t_{32}t_{21}\, S_3^+ - \phi_3^{-1} t_{32}t_{21}\, S_3^-}{\phi_3^{-1}\big(r_{32}r_{21}\phi_2 + \phi_2^{-1}\big) - \phi_3 r_{34}\big(r_{21}\phi_2 + r_{32}\phi_2^{-1}\big)}. \tag{69}$$

The last step is to obtain the source terms of Eq. (9). We first evaluate

$$\begin{pmatrix} \phi_3^{-1} & 0 \\ 0 & \phi_3 \end{pmatrix} \begin{pmatrix} \frac{1}{t_{3s}} & \frac{r_{3s}}{t_{3s}} \\ \frac{r_{3s}}{t_{3s}} & \frac{1}{t_{3s}} \end{pmatrix} \begin{pmatrix} \phi_s & 0 \\ 0 & \phi_s^{-1} \end{pmatrix} = \frac{1}{t_{3s}} \begin{pmatrix} \phi_3^{-1}\phi_s & \phi_3^{-1} r_{3s}\phi_s^{-1} \\ \phi_3 r_{3s}\phi_s & \phi_3\phi_s^{-1} \end{pmatrix}. \tag{70}$$

Therefore,

$$\begin{aligned} \begin{pmatrix} S_3^+ \\ S_3^- \end{pmatrix} &= \frac{1}{t_{3s}} \left[\begin{pmatrix} \phi_3^{-1}\phi_s & \phi_3^{-1} r_{3s}\phi_s^{-1} \\ \phi_3 r_{3s}\phi_s & \phi_3\phi_s^{-1} \end{pmatrix} - \begin{pmatrix} 1 & r_{3s} \\ r_{3s} & 1 \end{pmatrix} \right] \mathbf{E}_s \\ &= \frac{1}{t_{3s}} \begin{pmatrix} \phi_3^{-1}\phi_s - 1 & r_{3s}\big(\phi_3^{-1}\phi_s^{-1} - 1\big) \\ r_{3s}(\phi_3\phi_s - 1) & \phi_3\phi_s^{-1} - 1 \end{pmatrix} \mathbf{E}_s. \end{aligned} \tag{71}$$

$$S_3^+ = \frac{1}{t_{3s}} \Big[(\phi_3^{-1}\phi_s - 1)E_s^+ + r_{3s}(\phi_3^{-1}\phi_s^{-1} - 1)E_s^- \Big], \tag{72}$$

$$S_3^- = \frac{1}{t_{3s}} \Big[r_{3s}(\phi_3\phi_s - 1)E_s^+ + (\phi_3\phi_s^{-1} - 1)E_s^- \Big]. \tag{73}$$

In the case of second-harmonic generation (SHG), two source refractive indices must be considered, namely $n_s = n_3(\omega)$ and $n_s = 0$. Consequently, two distinct phase factors,

$$\phi_s = \exp[i\, 2\, n_s\, k_0\, d_3]\,, \tag{74}$$

and their corresponding source fields contribute to the nonlinear response. The amplitudes of the source fields depend on the chosen value of n_s as well as on the square of the fundamental fields. Accordingly, the source-field amplitudes in layer 3 are given by

$$\begin{pmatrix} E_s^+ \\ E_s^- \end{pmatrix} = \frac{\epsilon_0 \chi^{(2)}}{\epsilon_0\big[(n_s)^2 - \epsilon_3(2\omega)\big]} \begin{pmatrix} |E_3^+(\omega)|^2 \\ |E_3^-(\omega)|^2 \end{pmatrix}. \tag{75}$$

Here, $|E_3^+(\omega)|^2$ and $|E_3^-(\omega)|^2$ denote the squared amplitudes of the forward- and backward-propagating fundamental fields in layer 3, respectively, obtained from the linear transfer-matrix formalism described by Eq. (52).

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[1] D. S. Bethune, "Optical Harmonic Generation and Mixing in Multilayer Media: Analysis Using Optical Transfer Matrix Techniques," *Journal of the Optical Society of America B*, **6**, 910–916 (1989).